

27-06-2026

Agenda:

- kNN — covered (FE) •
- cross-validation

kNN → K-nearest Neighbours
↓
no of neighbours → supervised algo
→ classification
→ regression

→ Instead of relying on eqn why not use the data itself to make decision.

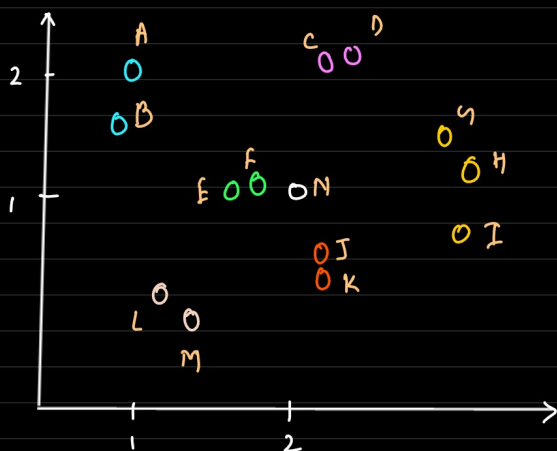
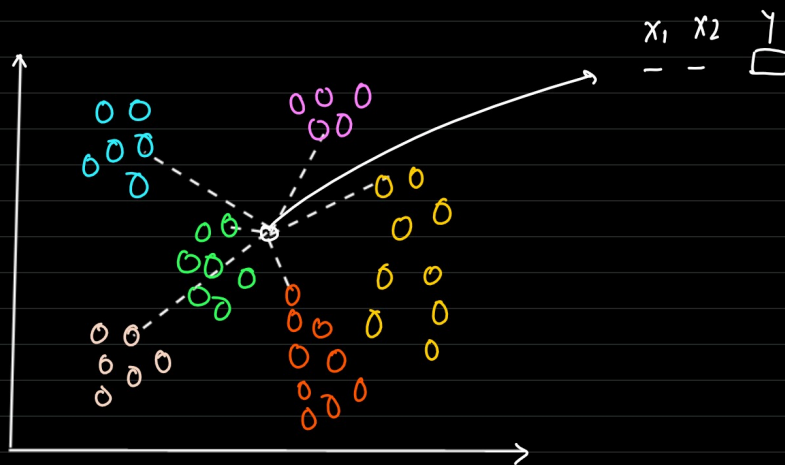
→ we don't compute any parameter (α, β, γ)

→ no mathematical eqn of line or curve

use for case:

data doesn't form a nice line

→ called a lazy learner, because it doesn't learn any parameters.



distance (Euclidean)

N - A
(2, 1) (1, 2)

N - B

N - C

⋮

N - M

$$\sqrt{(2-1)^2 + (1-2)^2} = 1.414$$

Sort the distances in ascending order

Distance formula:

Euclidean distance: $\sqrt{\sum (x_i - x_j)^2}$

Manhattan distance: $\sum |x_i - x_j|$

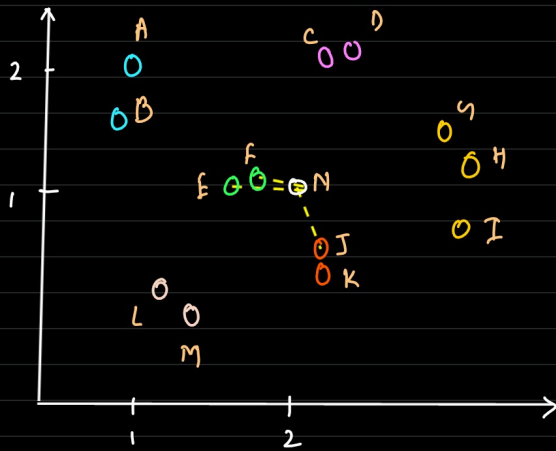
points closest to new point

$K \rightarrow$ select no of neighbours

$\rightarrow$ odd numbers

$\rightarrow 3$

$\rightarrow$ select 3 nearest point from the sorted list



$y \rightarrow$ salary

classification

$N \rightarrow (F, E, J)$

$\downarrow$

new point will be classified as Green.

Regression

$N \rightarrow (F, E, J)$

(x_1, x_2, x_3) (x_1, x_2, x_3) (x_1, x_2, x_3)

$\downarrow$ $\downarrow$ $\downarrow$

avg $\rightarrow$ salary

optimal value of K :

$\rightarrow$

$K \rightarrow 1 \rightarrow P, R$

$K \rightarrow 2 \rightarrow$

$K \rightarrow 3 \rightarrow$

Advantage:

- very simple
- easy to decode input

Balanced ($K = 5, 7, 9$)

linear regression $\rightarrow$ save $\rightarrow$ obj 1

$\downarrow$

$\beta/w, c/m, c$

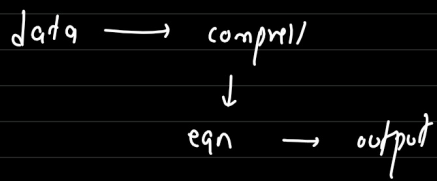
logistic regression $\rightarrow$ save $\rightarrow$ obj 2

KNN $\rightarrow$ save $\rightarrow$ obj 3

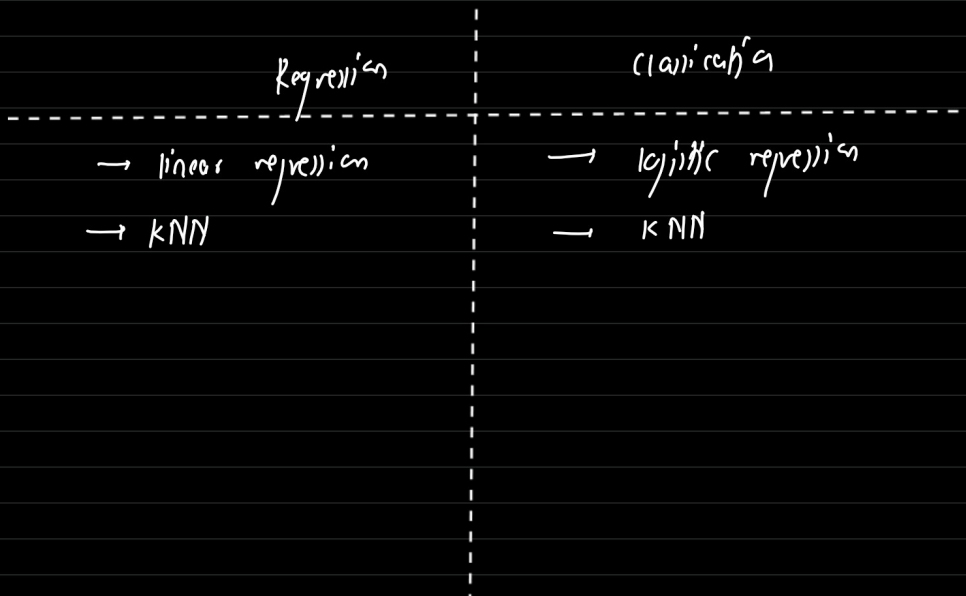
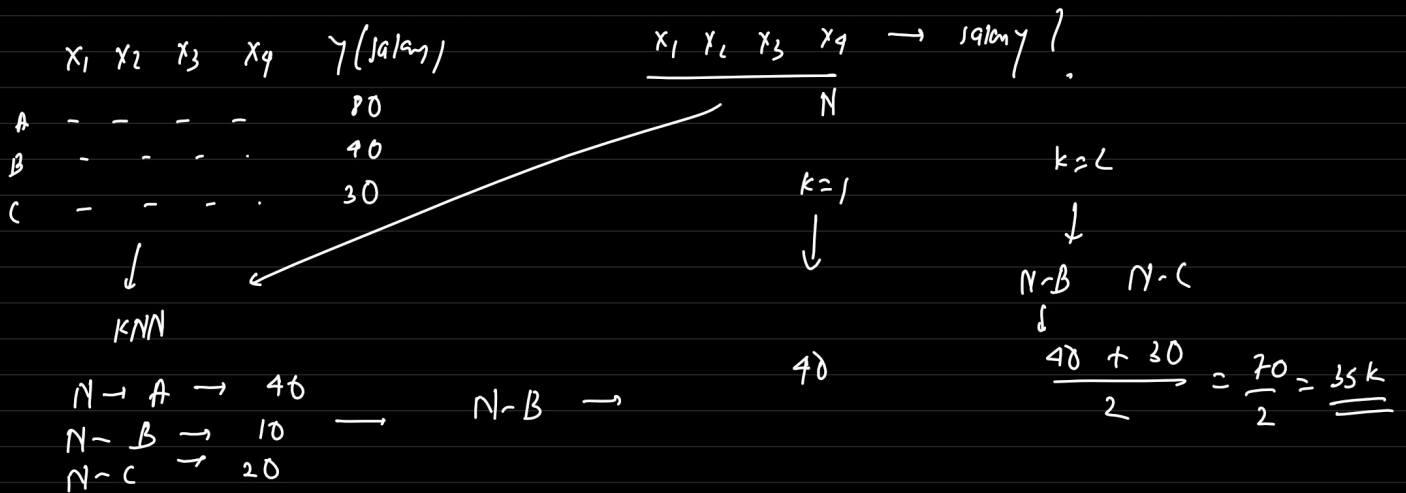
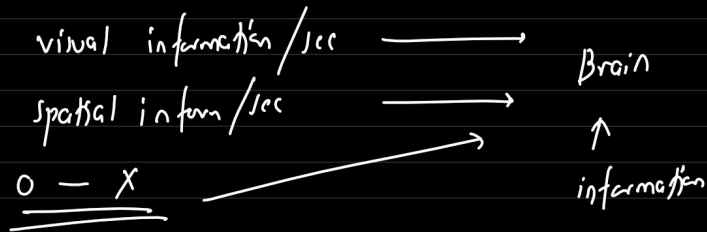
$\downarrow$ data $\rightarrow$ data

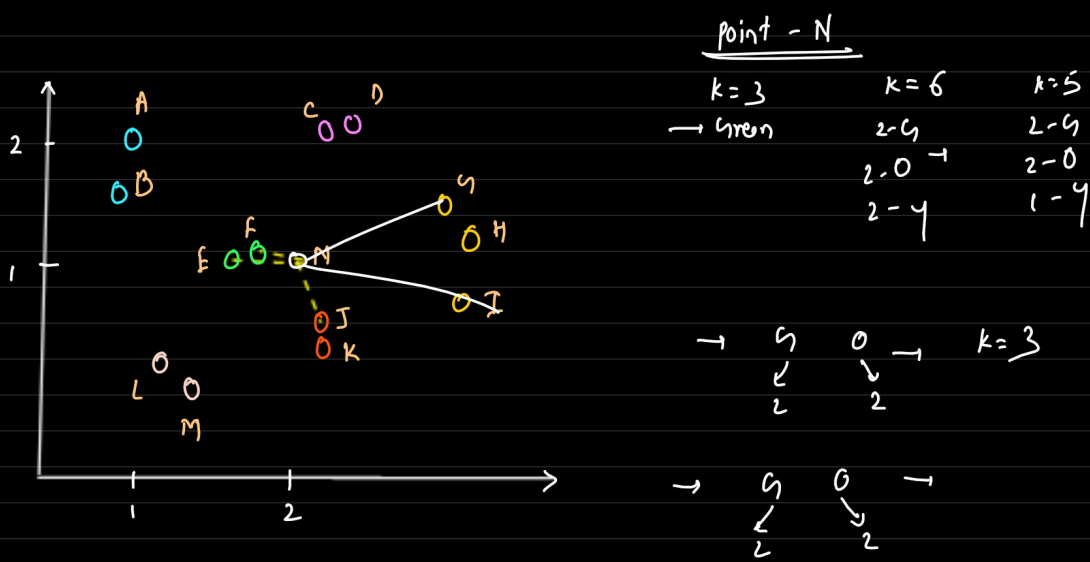
1 million data

input $\rightarrow$ 1 million distance calculation $\rightarrow$ output

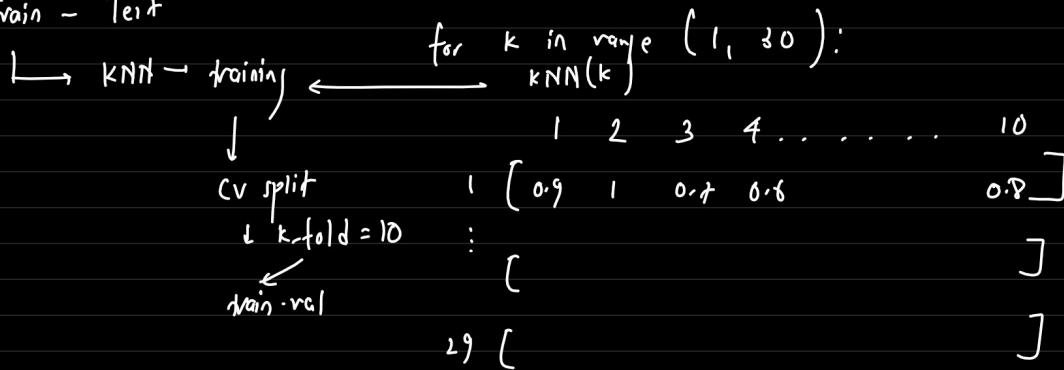


wiki, reddit, twitter, $\rightarrow$ chatgpt $\leftarrow$
 github





Train - Test



avg $\rightarrow k=1 \rightarrow$

$\max(avg) \rightarrow \text{index} \rightarrow k$

avg $\rightarrow k=2 \rightarrow$

$\rightarrow \max(avg) \rightarrow \text{max value}$
 $\rightarrow \text{avgmax} \rightarrow \text{index of max value}$

avg	k
	1
	2
	3

K_range[np.argmax(cv_scores)]

→ [0 → 9]

